

MATH40003 - Linear Algebra and Groups

Based on lectures by Prof. T. Schedler and Dr M. Zordan
Notes taken by

Autumn 2025 – Spring 2026
Compiled on May 9, 2026

These notes are not official and do not represent the content of lectures or classes.

Be mindful of any mistakes or omissions if using these notes for self-study or
revision. Any and all errors are my own.

More details at <https://stones-y.github.io>.

Contents

Introduction	2
1 Fields and Vector Spaces	3
1.1 Fields	3
1.2 Vector Spaces	4
2 Matrices	7
2.1 Matrices and Linear Maps	7
2.2 Matrix Multiplication	8
2.2.1 Motivation	8
2.2.2 Some more formal properties of matrices	10
2.2.3 Special Matrices	12

Introduction

Why do we study linear algebra?

- 2D and 3D Geometry
- Signal processing: audio and video samples, data and communications
- and many more!

Chapter 1

Fields and Vector Spaces

1.1 Fields

Definition 1.1 (Field). Let F be a set with two binary operations, addition $+: F \times F \rightarrow F$ and multiplication $\cdot: F \times F \rightarrow F$. We say that F is a *field* if it satisfies the following axioms for all $a, b, c \in F$:

Axioms for Addition $(F, +)$:

A1. Associativity: $(a + b) + c = a + (b + c)$

A2. Commutativity: $a + b = b + a$

A3. Additive Identity: There exists an element $0 \in F$ such that $a + 0 = a$.

A4. Additive Inverse: For each $a \in F$, there exists an element $-a \in F$ such that $a + (-a) = 0$.

Axioms for Multiplication $(F, \cdot)$:

M1. Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

M2. Commutativity: $a \cdot b = b \cdot a$

M3. Multiplicative Identity: There exists an element $1 \in F$, with $1 \neq 0$, such that $a \cdot 1 = a$.

M4. Multiplicative Inverse: For each $a \in F \setminus \{0\}$ there exists an element $a^{-1} \in F \setminus \{0\}$ such that $a \cdot a^{-1} = 1$.

Distributivity:

D1. $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

Remark. This is all the structure we need from the real numbers to do linear algebra. We can do linear algebra with many other fields, not just $\mathbb{R}$, like the finite field $\mathbb{F}_2$. Abstracting the scalars into an arbitrary field F allows for some rich and interesting results in linear algebra.

Example 1.2. The integers modulo p , $\mathbb{F}_p := \{0, 1, \dots, p-1\}$, where p is prime, form a field. Addition and multiplication are defined modulo p . All properties except the existence of a multiplicative inverse follow easily from the fact that $\mathbb{Z}$ is a commutative ring.

Lemma 1.3. In $\mathbb{F}_p$, $xy = 0 \iff x = 0$ or $y = 0$.

Proof. $xy = 0 \implies p \mid xy \implies p \mid x$ or $p \mid y$ (by Euclid's lemma). This means $x \equiv 0 \pmod{p}$ or $y \equiv 0 \pmod{p}$. $\square$

Theorem 1.4. For every $a \in \mathbb{F}_p \setminus \{0\}$ there exists $a^{-1} \in \mathbb{F}_p \setminus \{0\}$ such that $aa^{-1} = 1 = a^{-1}a$.

Proof. Given $a \in \mathbb{F}_p$, by commutativity we only need to find a $b \in \mathbb{F}_p$ such that $ab = 1$. Finding an explicit construction for b is difficult, so we instead show that the set of multiples $S := \{a \cdot 0, a \cdot 1, \dots, a \cdot (p-1)\}$ is exactly equal to $\mathbb{F}_p$.

It suffices to show that the p elements $a \cdot 0, a \cdot 1, \dots, a \cdot (p-1) \in \mathbb{F}_p$ are all distinct. Suppose $a \cdot c = a \cdot d$. Then $a \cdot (c - d) = 0$.

By our lemma, since $a \neq 0$ ($p \nmid a$), we must have $c - d = 0$, which implies $c = d$. Thus, multiplication by a maps distinct elements to distinct elements. Since there are p elements in $\mathbb{F}_p$, the set S covers all of $\mathbb{F}_p$. Because $1 \in \mathbb{F}_p$, there must be some element $b \in \{0, \dots, p-1\}$ such that $ab = 1$. $\square$

Corollary 1.5. $\mathbb{F}_p$ is a field.

Remark. The proof above relies on the Pigeonhole Principle. Because multiplication by a non-zero element a is injective on the finite set $\mathbb{F}_p$, it must also be surjective.

Definition 1.6 (Subfield). A subset $F \subseteq E$ of a field E is a *subfield* if $1 \in F$ and it is closed under arithmetic: whenever $a, b \in F$ we have $a + b, ab \in F$; if $a \in F$ then $-a \in F$; and if $a \in F \setminus \{0\}$ then $a^{-1} \in F$.

Remark. This definition is slightly redundant. To prove a subset $F \subseteq E$ is a subfield, it actually suffices to show closure under addition and multiplication, the existence of multiplicative inverses for non-zero elements, and that $-1 \in F$. Because if $-1 \in F$ and we have the closures above, then $(-1)(-1) = 1 \in F$ and $(-1) + 1 = 0 \in F$. Additive inverses follow from closure of addition with -1 .

Example 1.7. Some examples of subfields are:

- $\mathbb{Q}[i] := \{a + bi : a, b \in \mathbb{Q}\} \subseteq \mathbb{C}$ is a field (the "Gaussian rationals"). It is a subfield of $\mathbb{C}$.
- $F[X] = \{a_0 + a_1X + \dots + a_nX^n : a_i \in F\}$ is *not* a field because non-constant polynomials lack multiplicative inverses (e.g., $1/X \notin F[X]$). However, it is a vector space.
- The field of rational functions $F(X) := \left\{ \frac{p(X)}{q(X)} : p, q \in F[X], q \neq 0 \right\}$ is a field.

1.2 Vector Spaces

Definition 1.8. The vector space F^n of n -tuples over a field F is defined as:

$$F^n := \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} : x_1, \dots, x_n \in F \right\}$$

Remark. This is the primordial example of a vector space. Every finite-dimensional vector space is structurally identical (isomorphic) to F^n . The geometric intuition of $\mathbb{R}^n$ (arrows, scaling, and adding tip-to-tail) carries over algebraically to arbitrary fields.

Definition 1.9 (Commutative Group). A commutative group $(V, +)$ is a set equipped with a binary operation $+: V \times V \rightarrow V$ satisfying:

- **Associativity:** For all $u, v, w \in V$, $u + (v + w) = (u + v) + w$.
- **Commutativity:** For all $v, w \in V$, $v + w = w + v$.
- **Additive Identity:** There exists $0 := (0, \dots, 0) \in V$ such that for all $v \in V$, $v + 0 = 0 + v = v$.
- **Additive Inverse:** For all $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0 = (-v) + v$.

Lemma 1.10. In a commutative group $(V, +)$, the zero vector and additive inverses are unique.

Proof. Given another zero vector $0'$, we have $0' = 0 + 0' = 0$. Thus, the zero vector is unique.

If w also acts as an additive inverse to v (so $w + v = 0$), then:

$$w = w + 0 = w + (v + (-v)) = (w + v) + (-v) = 0 + (-v) = -v.$$

Thus, the additive inverse is unique. □

Definition 1.11 (Vector Space). A vector space V over a field F is a commutative group $(V, +)$ together with an action $F \times V \rightarrow V$, mapped by $(\lambda, v) \mapsto \lambda v$, called *scalar multiplication*, satisfying the following properties for all $\lambda, \mu \in F$ and $v, w \in V$:

- **Associativity:** $\lambda(\mu v) = (\lambda\mu)v$
- **Distributivity of scalar sums:** $(\lambda + \mu)v = \lambda v + \mu v$
- **Distributivity of vector sums:** $\lambda(v + w) = \lambda v + \lambda w$
- **Unit multiplication:** $1 \cdot v = v$

Remark. Structurally, one can think of a vector space as having two tiers: the “ground” field F provides the scalars (stretching factors), while the commutative group V provides the vectors (the objects being stretched and added; indeed, F acts on V to get scalar multiples of vectors). The axioms ensure these tiers interact in a linear way.

Example 1.12. Other examples of vector spaces include:

- $F[X] = \{a_0 + a_1X + \cdots + a_nX^n : a_i \in F\}$ (polynomials).
- Subfields $F \subseteq E$ (E acts as a vector space over its subfield F).
- $\text{Fun}(X, F) = \{f : X \rightarrow F\}$ (functions from an arbitrary set X into F).

Chapter 2

Matrices

2.1 Matrices and Linear Maps

Matrices are the "coordinated" way to describe linear transformations. We will formalise this in Theorem 2.3 below. But first, we define a linear transformation.

Definition 2.1 (Linear transformation). Fix a field F and two vector spaces V, W over F . A *linear transformation* (or linear map) is a function $T : V \rightarrow W$ such that:

- **Additivity:** $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$.
- **Homogeneity:** $T(\lambda v) = \lambda T(v)$ for all $\lambda \in F$ and $v \in V$.

Lemma 2.2. *If $T : V \rightarrow W$ is a linear transformation, then $T(0) = 0$.*

Proof. By linearity, $T(0) = T(0 + 0) = T(0) + T(0)$. Adding $-T(0)$ (the additive inverse of $T(0)$ in W) to both sides yields $T(0) = 0$. $\square$

Recall that any finite-dimensional vector space V , with dimension n , is isomorphic to F^n . For now, when we say we have written V and W in coordinates, we mean we replace V and W with F^n and F^m . This motivates the concept of matrices.

Theorem 2.3. *For all linear maps $T : F^n \rightarrow F^m$ defined by $(x_1, \dots, x_n) \mapsto (y_1, \dots, y_m)$, there exists an array of numbers (a_{ij}) such that:*

$$y_i = \sum_{j=1}^n a_{ij} x_j$$

Proof. We define the standard elementary vectors $e_j \in F^n$ to have value 1 in the j -th entry, and 0 everywhere else. Since $T(e_j)$ is a vector in F^m , we can write it as a list of m coordinates. For all j , define $a_{1j}, \dots, a_{mj} \in F$ by:

$$T(e_j) = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}$$

(Idea: think of $T(e_j)$ as the j^{th} column of the matrix (a_{ij})).

Any input vector can be written as $(x_1, \dots, x_n) = x_1 e_1 + \dots + x_n e_n$ by our definition of addition in F^n . By the linearity of T , we have:

$$\begin{aligned} T(x_1, \dots, x_n) &= T(x_1 e_1 + \dots + x_n e_n) \\ &= x_1 T(e_1) + \dots + x_n T(e_n) \\ &= \sum_{j=1}^n x_j \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} \end{aligned}$$

Taking the i^{th} row (coordinate) of the above sum, we get:

$$y_i = \sum_{j=1}^n a_{ij} x_j$$

as required. □

Remark. The converse is also true: any function $T : F^n \rightarrow F^m$ given by such an array (a_{ij}) is a linear map. This is fundamental: *linear maps are exactly the functions that can be represented by matrices.*

Observe that all the information needed to compute this linear transformation is encoded in the array $A := (a_{ij})$.

Definition 2.4 (Matrix). The set of $m \times n$ matrices over a field F is defined as:

$$M_{m \times n}(F) := \left\{ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} : a_{ij} \in F \right\}$$

How do we conveniently encode the action of T using this? We define matrix-vector multiplication:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

The upshot is that, given a linear map $T : F^n \rightarrow F^m$, we *define* matrix multiplication with a column vector precisely to execute this map.

2.2 Matrix Multiplication

2.2.1 Motivation

First, we motivate the definition of general matrix multiplication. Consider two linear maps:

- $T : F^k \rightarrow F^n$ represented by the matrix $B \in M_{n \times k}(F)$
- $S : F^n \rightarrow F^m$ represented by the matrix $A \in M_{m \times n}(F)$

Define the composite linear map $R : F^k \rightarrow F^m$ by $R = S \circ T$. By Claim $(\star)$, this new linear map must also have a matrix representation, which we will call $C \in M_{m \times k}(F)$.

We want to calculate $c_{i\ell}$, the $(i, \ell)^{\text{th}}$ element of the matrix C . Let's calculate the ℓ^{th} column of C , call it C_ℓ . We know that each column of C represents the effect of the linear map R on the standard basis vector e_ℓ :

$$C_\ell = R(e_\ell) = S(T(e_\ell))$$

We know $T(e_\ell)$ is exactly the ℓ^{th} column of B , call it B_ℓ . So we must calculate $S(B_\ell)$. Because B_ℓ is a column vector in F^n , evaluating $S(B_\ell)$ is equivalent to taking a linear combination of the columns of A (which we will denote $A_1, \dots, A_n$):

$$S(B_\ell) = b_{1\ell}A_1 + \dots + b_{n\ell}A_n$$

To find $c_{i\ell}$, we take the i^{th} row of this result:

$$\begin{aligned} c_{i\ell} &= (C_\ell)_i = b_{1\ell}(A_1)_i + \dots + b_{n\ell}(A_n)_i \\ &= b_{1\ell}a_{i1} + \dots + b_{n\ell}a_{in} \\ &= \sum_{j=1}^n a_{ij}b_{j\ell} \end{aligned}$$

This is exactly the formula for $c_{i\ell}$ that we use to define matrix multiplication. Therefore, the fact that composition of linear maps gives the formula above for the numbers $c_{i\ell}$ is why we define matrix multiplication in the following way:

Definition 2.5 (Matrix operations).

- Matrix multiplication is the function $\cdot : M_{m \times n}(F) \times M_{n \times p}(F) \rightarrow M_{m \times p}(F)$ defined by:

$$(a_{ij})_{m \times n} (b_{k\ell})_{n \times p} := \left(c_{i\ell} = \sum_{j=1}^n a_{ij}b_{j\ell} \right)_{m \times p}$$

- Matrix addition is the function $+$: $M_{m \times n}(F) \times M_{m \times n}(F) \rightarrow M_{m \times n}(F)$ defined by:

$$(a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij}).$$

- Scalar multiplication for matrices is the function $\cdot : F \times M_{m \times n}(F) \rightarrow M_{m \times n}(F)$ defined by:

$$\lambda(a_{ij}) = (\lambda a_{ij}).$$

Remark. Just like $F^n \cong M_{n \times 1}(F)$, the set $M_{m \times n}(F)$ forms a vector space under matrix addition and scalar multiplication. However, matrix multiplication gives $M_{n \times n}(F)$ additional structure beyond just a vector space (it forms an algebra).

Definition 2.6 (Square Matrix). A matrix is square if $A \in M_{n \times n}(F)$ (the number of rows equals the number of columns). Square matrices of a fixed size can be multiplied by each other.

Proposition 2.7. *Matrix multiplication is associative: $(AB)C = A(BC)$.*

Proof.

$$\begin{aligned} ((AB)C)_{i\ell} &= \sum_j (AB)_{ij} c_{j\ell} = \sum_j \left(\sum_k a_{ik} b_{kj} \right) c_{j\ell} = \sum_{j,k} a_{ik} b_{kj} c_{j\ell} \\ (A(BC))_{i\ell} &= \sum_k a_{ik} (BC)_{k\ell} = \sum_k a_{ik} \left(\sum_j b_{kj} c_{j\ell} \right) = \sum_{j,k} a_{ik} b_{kj} c_{j\ell} \end{aligned}$$

Therefore, $((AB)C)_{i\ell} = (A(BC))_{i\ell}$. $\square$

Corollary 2.8. *Given linear maps $S : F^m \rightarrow F^n$ and $T : F^n \rightarrow F^p$ with matrix representations A and B (so $S(v) = Av$ and $T(v) = Bv$), the matrix representation of the composite map $T \circ S : F^m \rightarrow F^p$ is given by the matrix product BA .*

Proof. $T \circ S(v) = T(S(v)) = B(Av) = (BA)v$ by associativity. $\square$

2.2.2 Some more formal properties of matrices

Definition 2.9 (Bilinear Map). Let V, W, U be vector spaces over F . A map $f : V \times W \rightarrow U$ mapping $(v, w) \mapsto f(v, w)$ is *bilinear* if it is linear in each argument when the other is held fixed:

- $(\lambda_1 v_1 + \lambda_2 v_2, w) \mapsto \lambda_1 f(v_1, w) + \lambda_2 f(v_2, w)$
- $(v, \lambda_1 w_1 + \lambda_2 w_2) \mapsto \lambda_1 f(v, w_1) + \lambda_2 f(v, w_2)$

Remark. Note that this implies the standard scalar extraction property $f(\lambda v, w) = f(v, \lambda w) = \lambda f(v, w)$ (simply set $\lambda_2 = 0$).

Example 2.10. The dot product $F^n \times F^n \rightarrow F$, defined by $((a_1, \dots, a_n), (b_1, \dots, b_n)) \mapsto a_1 b_1 + \dots + a_n b_n$ is bilinear. This follows easily from the distributive properties of the field F .

We want to prove that matrix multiplication is bilinear. To do this, we first establish how matrix multiplication interacts with columns and rows.

Proposition 2.11. *Let $A \in M_{m \times n}(F)$ and $B = (B_1, \dots, B_p) \in M_{n \times p}(F)$, where B_j are the columns of B . Then:*

$$AB = (AB_1, \dots, AB_p)$$

Proof. By the formula, $(AB)_{i\ell} = \sum_k a_{ik} b_{k\ell}$. Notice that $\sum_k a_{ik} b_{k\ell}$ is exactly the i^{th} element of the matrix-vector product AB_ℓ . Thus, the ℓ^{th} column of AB is exactly the vector AB_ℓ . $\square$

Remark. This provides a *column-wise* way to calculate matrix products. To find the entire j^{th} column of AB , just calculate the matrix-vector product AB_j . The j^{th} column of AB **only** depends on the j^{th} column of B .

Proposition 2.12. *We can separate rows under right multiplication. Let B_i be the rows of B . Then:*

$$\begin{pmatrix} B_1 \\ \vdots \\ B_p \end{pmatrix} A = \begin{pmatrix} B_1 A \\ \vdots \\ B_p A \end{pmatrix}$$

Proof. Same logic as above, switching indices appropriately. $\square$

Corollary 2.13. *Matrix multiplication is bilinear.*

Proof. We prove linearity in the second argument. Let A be an $m \times n$ matrix, B and C be $n \times p$ matrices, and $\lambda \in F$.

$$\begin{aligned} A(\lambda B) &= A(\lambda B_1 \dots \lambda B_n) \\ &= (A(\lambda B_1) \dots A(\lambda B_n)) && \text{(Column separation)} \\ &= (\lambda(AB_1) \dots \lambda(AB_n)) && \text{(Matrix-vector linearity)} \\ &= \lambda(AB_1 \dots AB_n) \\ &= \lambda(AB) \end{aligned}$$

And for addition:

$$\begin{aligned} A(B + C) &= A(B_1 + C_1 \dots B_n + C_n) \\ &= (A(B_1 + C_1) \dots A(B_n + C_n)) \\ &= (A(B_1) + A(C_1) \dots A(B_n) + A(C_n)) \\ &= (A(B_1) \dots A(B_n)) + (A(C_1) \dots A(C_n)) \\ &= AB + AC \end{aligned}$$

Linearity in the first argument follows similarly. $\square$

Definition 2.14 (Transpose). The transpose of a matrix $A = (a_{ij}) \in M_{m \times n}(F)$ is defined as $A^t = (a_{ji}) \in M_{n \times m}(F)$.

Proposition 2.15. *The transpose operation satisfies the following properties:*

- *Transpose is linear:* $(A + B)^t = A^t + B^t$ and $(\lambda A)^t = \lambda A^t$.
- $(A^t)^t = A$.
- $(AB)^t = B^t A^t$.

Proof. We prove the product rule. Let $C = AB$. Then $c_{ij} = \sum_k a_{ik} b_{kj}$. The (i, j) entry of $(AB)^t$ is the (j, i) entry of AB , which is $\sum_k a_{jk} b_{ki}$. The (i, j) entry of $B^t A^t$ is the dot product of the i^{th} row of B^t and the j^{th} column of A^t . This is exactly $\sum_k (B^t)_{ik} (A^t)_{kj} = \sum_k b_{ki} a_{jk} = \sum_k a_{jk} b_{ki}$. $\square$

Remark. Notice that the order reverses in $(AB)^t = B^t A^t$. This is necessary for the dimensions to align. For example, if A is 1×5 and B is 5×1 , AB is 1×1 . A^t is 5×1 and B^t is 1×5 . The product $A^t B^t$ is impossible, but $B^t A^t$ results in a 1×1 matrix.

2.2.3 Special Matrices

Definition 2.16 (Symmetric Matrix). A square matrix A is *symmetric* if $A = A^t$.

Definition 2.17 (Orthogonal Matrix). A square matrix A is *orthogonal* if $AA^t = I$.

Remark. For a square matrix, $AA^t = I \iff A^t A = I$. What does it mean for $AA^t = I$? If A_i are the rows of A , then $(AA^t)_{ij} = A_i \cdot A_j$. Therefore:

$$A_i \cdot A_j = \delta_{ij} := \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

This means the *rows* of A are orthonormal. Similarly, $A^t A = I$ implies the *columns* of A are orthonormal. Geometrically in $\mathbb{R}^n$, multiplying by an orthogonal matrix represents a rigid transformation (a rotation or reflection) that preserves lengths and angles.

Definition 2.18 (Triangular and Diagonal Matrices). Let $A = (a_{ij})$ be a matrix.

- A is *upper-triangular* if $a_{ij} = 0$ for all $i > j$.
- A is *lower-triangular* if $a_{ij} = 0$ for all $i < j$.
- A is *diagonal* if it is both upper and lower triangular (i.e., $a_{ij} = 0$ for all $i \neq j$).